

A

Fixed spectrum; different interactions

All 105 perfect matchings

$$f_M = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j$$

All 35 four-plus-four partitions

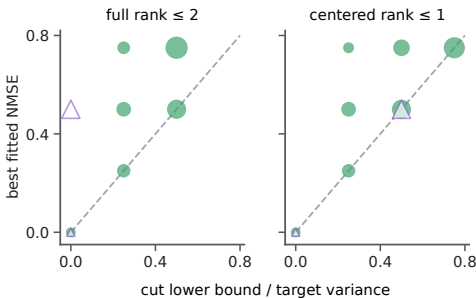
$$f_A = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right)$$

$$E[\nabla_x f \nabla_x f^T] = I_8/4; \text{rank} = 8$$

28 coefficients · 14 edges · root-only output

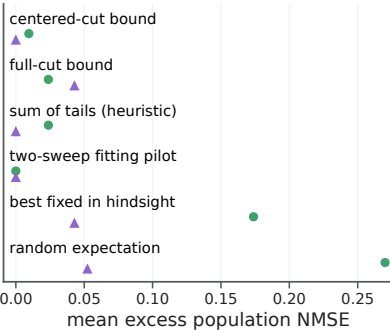
B

The centered constraint is stronger



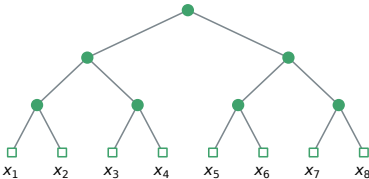
C

Selection within twelve candidates



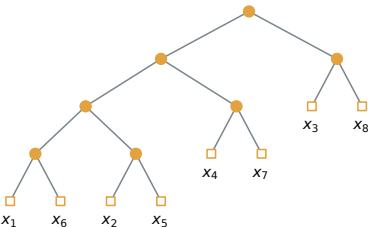
D

Parallel: minimum depth 3
root readout



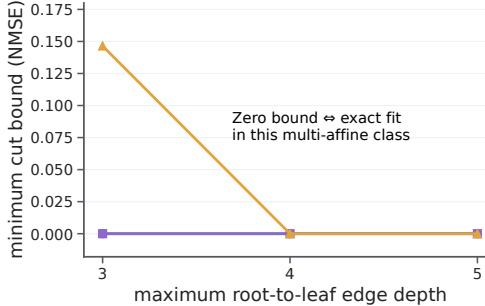
E

Nested: minimum depth 4
root readout



F

Depth constraint over all labeled trees



105 matching targets

35 quartic targets

24 nested controls

depth 3

depth 3

depth 4